CLASS 9 · CBSE · GANITA MANJARI PART 1 · FORMULA BANK

Chapter 6: Measuring Space: Perimeter and Area

Every formula and result of the chapter on one sheet

1	Circumference of a circle	$C = 2\pi r = \pi d$
2	Length of an arc	$\frac{\theta}{360} \times 2\pi r$
3	Perimeter of a sector	Arc length $+2r$
4	Perimeter of a semicircular region	$\pi r + 2r$
5	Area of a parallelogram	Base $\times$ height
6	Area of a triangle	$\frac{1}{2} \times$ base $\times$ height
7	Heron's formula	$\sqrt{s(s-a)(s-b)(s-c)}$, where $s = \frac{a+b+c}{2}$
8	Area of an equilateral triangle	$\frac{\sqrt{3}}{4}a^2$
9	Triangle, using the circumradius R	$\frac{abc}{4R}$
10	Triangle, using the inradius r	$r \times s$
11	Area of a trapezium	$\frac{1}{2}(a+b)h$
12	Area of a rhombus or kite	$\frac{1}{2}d_1d_2$
13	Brahmagupta's formula (cyclic quadrilateral)	$\sqrt{(s-a)(s-b)(s-c)(s-d)}$, where $s = \frac{a+b+c+d}{2}$
14	Area of a circle	πr^2
15	Area of a sector	$\frac{\theta}{360} \times \pi r^2$

16	Area of a minor segment	Area of the sector − area of the triangle
17	Squaring a rectangle	$ab = \left(\frac{a+b}{2}\right)^2 - \left(\frac{a-b}{2}\right)^2$

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